Quantum-Enhanced Deep Learning for Robust Night-Time License Plate Recognition

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Abstract

Automatic License Plate Recognition (ALPR) systems have achieved high accuracy in controlled daylight environments; however, their performance degrades significantly under night-time conditions characterized by low illumination, motion blur, and headlight glare. Current classical Deep Learning (DL) solutions often struggle to generalize across these noisy domains. This project proposes a **Hybrid Quantum-Classical Neural Network (HQCNN)** architecture that integrates classical deep feature extraction with Quantum Variational Circuits (VQC) for character recognition. By leveraging the high-dimensional feature mapping capabilities of quantum Hilbert spaces, this system aims to enhance the separability of character features in degraded images, offering a robust alternative to purely classical OCR systems for night-time surveillance.

1. Literature Review

Classical ALPR Evolution and Challenges The domain of Automatic License Plate Recognition (ALPR) has evolved significantly from early edge-detection-based methods to sophisticated deep learning architectures. Modern pipelines typically employ a two-stage approach comprising detection and recognition. State-of-the-art detectors, particularly the YOLO (You Only Look Once) family and SSD (Single Shot Multibox Detector), have become the industry standard for License Plate Detection (LPD) due to their ability to balance real-time inference speeds with high mean Average Precision (mAP) [1]. For the subsequent recognition stage, architectures such as CRNN (Convolutional Recurrent Neural Networks) and LPRNet have replaced traditional template matching. These models leverage Convolutional Neural Networks (CNNs) for spatial feature extraction and Recurrent Neural Networks (RNNs) or Connectionist Temporal Classification (CTC) for sequence decoding [2]. Despite these advancements, standard ALPR systems exhibit a marked decline in performance when deployed in unconstrained night-time environments. Research by *Laroca et al.* [3] highlights that adverse conditions—specifically low illumination, motion blur from high-speed vehicles, and "blooming" artifacts caused by headlight glare—severely degrade the edge and texture features that classical CNNs rely upon, often rendering plates unreadable.

Low-Light Image Enhancement Techniques To address the limitations of recognition in poor visibility, image enhancement is frequently integrated as a pre-processing step. Traditional signal processing techniques, such as Histogram Equalization (HE) and Gamma Correction, offer computationally inexpensive solutions but frequently introduce noise amplification and unnatural artifacts. Consequently, recent literature has shifted toward deep learning-based enhancement methods. Notable among these is Zero-DCE (Zero-Reference Deep Curve Estimation) [4], which frames enhancement as a curve estimation problem, allowing for dynamic range

adjustment without the need for paired low/normal light datasets. Similarly, Retinex-based deep networks attempt to decompose images into reflectance and illumination components to restore visibility. However, a significant gap remains: these enhancers are often trained for general perceptual quality rather than maximizing machine readability. When they hallucinate artifacts or over-smooth texture details, the downstream OCR accuracy suffers, necessitating a recognition model that is inherently more robust to residual noise.

Quantum Machine Learning in Vision Emerging research in Quantum Machine Learning (QML) offers a promising avenue for overcoming the limitations of classical feature extraction in noisy domains. Theoretical foundations established by *Schuld et al.* [5] and *Havlíček et al.* [6] suggest that Variational Quantum Classifiers (VQC) can leverage the high-dimensional Hilbert space of quantum mechanics to separate data classes that are inextricably overlapped in classical feature spaces. By utilizing quantum phenomena such as superposition and entanglement, these models can create highly complex decision boundaries with fewer distinct parameters. Recent hybrid implementations—integrating classical CNNs for dimensionality reduction with Quantum Convolutional layers or VQCs for classification—have demonstrated superior generalization on noisy benchmark datasets like MNIST and Fashion-MNIST [7]. Despite these theoretical and experimental successes, the application of hybrid quantum-classical networks to the specific, real-world problem of night-time License Plate Recognition remains unexplored. This project seeks to bridge this gap, proposing that a quantum-enhanced recognition layer can better preserve character identity amidst the severe pixel-level degradation typical of night-time surveillance.

2. Problem Statement

Existing ALPR systems exhibit a high failure rate (>30%) in night-time surveillance due to three specific degradation factors:

1. **Low Illumination:** Reduces contrast between characters and the plate background.
2. **Motion Blur:** Caused by fast-moving vehicles at night (requiring longer exposure times).
3. **Headlight Glare:** Creates overexposed regions that "wash out" character structural details.

There is a need for a recognition architecture that is not only robust to these pixel-level corruptions but also capable of learning more distinct decision boundaries for character classes than standard CNNs.

3. Project Objectives

1. **Develop a Pipeline:** To design a modular ALPR pipeline incorporating **YOLO-based detection**, **Zero-DCE image enhancement**, and a **Quantum-Enhanced OCR** module.
2. **Quantum Circuit Design:** To engineer a Parameterized Quantum Circuit (PQC) capable of processing latent feature vectors from a classical CNN and mapping them to character probability distributions.
3. **Benchmarking:** To quantitatively compare the proposed Hybrid HQCNN against state-of-the-art classical models (CRNN, LPRNet) on "Hard" subsets (night/blur) of the CCPD and UFPR-ALPR datasets.
4. **Ablation Study:** To analyze the specific contribution of the quantum layer to noise robustness.

4. Proposed Architecture (Methodology)

The system is designed as a sequential pipeline with three core stages:

Stage 1: Detection & Pre-processing

- **Input:** Raw CCTV video frames (Night-time).
- **Detection: YOLOv8** (finetuned) localizes the license plate bounding box.
- **Enhancement:** The cropped plate is passed through a **Zero-DCE** module. This lightweight network estimates pixel-wise adjustment curves to restore brightness and suppress glare without adding significant latency.

Stage 2: Feature Extraction (Classical)

- **Backbone:** A lightweight CNN (e.g., **ResNet-18** or a custom 4-layer CNN) processes the enhanced image.
- **Dimensionality Reduction:** The final convolutional feature maps are flattened and passed through a Fully Connected (FC) layer to reduce the dimensionality to N features (where N corresponds to the number of qubits available, e.g., 4, 8, or 10 qubits).

Stage 3: Quantum Character Recognition (The Novelty)

Instead of a standard Softmax classifier, we utilize a **Variational Quantum Circuit (VQC)**:

1. State Preparation (Encoding): The classical feature vector $\mathbf{x}$ is encoded into the quantum state using Angle Encoding (rotation gates R_x, R_y).

   $$|\psi(\mathbf{x})\rangle = U_{enc}(\mathbf{x}) |0\rangle^{\otimes N}$$
2. **Ansatz (Process):** A trainable sequence of entanglement gates (CNOT) and rotation gates parameterized by weights θ. This layer allows the model to learn correlations between features in quantum space.
3. **Measurement:** Expectation values are measured on the Z-basis to obtain the output probability distribution for the character class (0-9, A-Z).

5. Datasets and Experimental Setup

5.1 Primary Datasets

- **CCPD (Chinese City Parking Dataset):** Contains over 200k images with diverse conditions. We will specifically use the CCPD-Night, CCPD-Blur, and CCPD-Challenge subsets.
- **UFPR-ALPR:** A high-resolution dataset focused on Brazilian plates, excellent for testing OCR on non-standardized formats.

5.2 Evaluation Strategy

We will create a "Degradation Benchmark" by artificially augmenting the clean datasets with:

- Gaussian Noise (simulating high-ISO grain).
- Motion Blur kernels (simulating speed).
- Local overexposure masks (simulating headlight glare).

Metrics:

- **Character Accuracy (CA):** % of correctly recognized characters.
- **Full Sequence Accuracy (FSA):** % of plates where *every* character is correct.
- **Frames Per Second (FPS):** Inference speed (simulation mode).

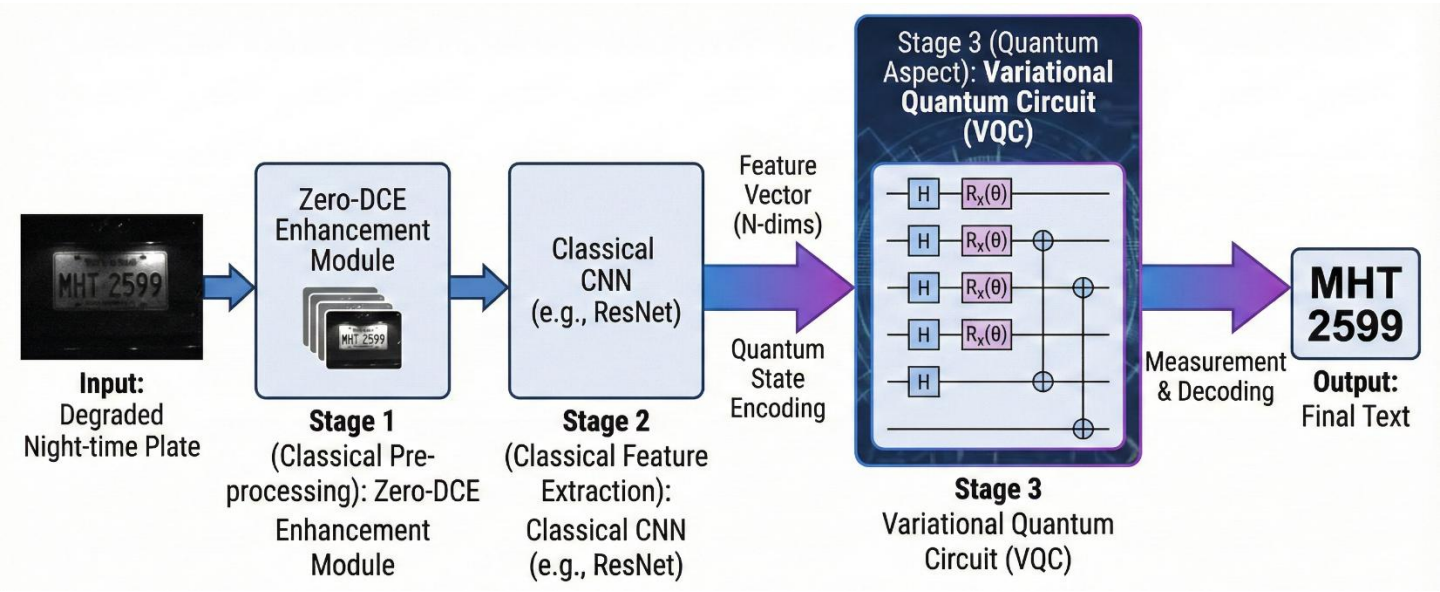

*Figure 1: **Proposed Hybrid Quantum-Classical Pipeline.** The system processes raw night-time imagery through classical enhancement and feature extraction before mapping latent features into a high-dimensional quantum Hilbert space via a parameterized Variational Quantum Circuit for robust character classification.*

6. Expected Outcomes

- A functional prototype of a Hybrid Quantum-Classical OCR system.
- Empirical evidence showing that the Quantum layer provides a >5% accuracy improvement over classical baselines on "severe noise" datasets.
- A research paper targeting conferences like **ICPR**, **CVPR (Workshops)**, or journals like **IEEE Transactions on Intelligent Transportation Systems**.

7. References

[1] Bochkovskiy, A., Wang, C. Y., & Liao, H. Y. M. (2020). *YOLOv4: Optimal speed and accuracy of object detection*. arXiv preprint arXiv:2004.10934.

[2] Zherzdev, E., & Gruzdev, A. (2018). *LPRNet: License Plate Recognition via Deep Neural Networks*. arXiv preprint.

[3] Laroca, R., et al. (2021). *A Robust Real-Time Automatic License Plate Recognition Based on the YOLO Detector*. IEEE Transactions on Intelligent Transportation Systems.

[4] Guo, C., et al. (2020). *Zero-Reference Deep Curve Estimation for Low-Light Image Enhancement*. CVPR.

[5] Schuld, M., Sinayskiy, I., & Petruccione, F. (2015). *An Introduction to Quantum Machine Learning*. Contemporary Physics.

[6] Havlíček, V., et al. (2019). *Supervised Learning with Quantum-Enhanced Feature Spaces*. Nature.

[7] Chen, S. Y. C., et al. (2021). *Quantum Convolutional Neural Networks*. Nature Physics (Related methodology).

[8] PennyLane: A Cross-Platform Python Library for Quantum Machine Learning, Xanadu, 2020.